



Time: 5 minutes

Outcome: Calculate Marginal Cost & Production-Cost Linkages.

Difficulty: Intermediate

Marginal Cost & Production-Cost Linkages

THE CORE IDEA

Marginal cost is the change in total cost divided by the change in output. If labor is the only variable input and its wage is fixed, divide the added worker's wage by that worker's marginal product. Lower marginal product then means higher cost per extra unit.

HOW TO RECOGNIZE IT

- Look for the change in total cost caused by producing one additional unit.
- Compute marginal cost as the change in total cost divided by the change in output.
- A fixed-cost change does not change marginal cost, but a variable-input cost change usually does.
- When marginal product falls, marginal cost tends to rise if the input price is unchanged.

WATCH OUT

Hold the wage and other input conditions fixed when linking marginal product to marginal cost. A fixed-cost change alone does not change the cost of an extra unit.

WORKED EXAMPLE: LINKING MARGINAL PRODUCT AND COST

Constant wage: \$120 per additional worker

	Marginal product	Added labor cost	Cost per extra unit
Fourth worker	12	\$120	\$10
Fifth worker	10	\$120	\$12

Each additional worker costs \$120. The fourth adds 12 units, so added cost per unit is $\$120 / 12 = \10 . The fifth adds 10 units, so it is $\$120 / 10 = \12 . With the wage unchanged, diminishing marginal product raises marginal cost.

CHECK YOURSELF

At a wage of \$120, another worker adds 8 units. What is the added labor cost per unit of output?